

Internal-External Boundary Attentions for Transparent Object Segmentation

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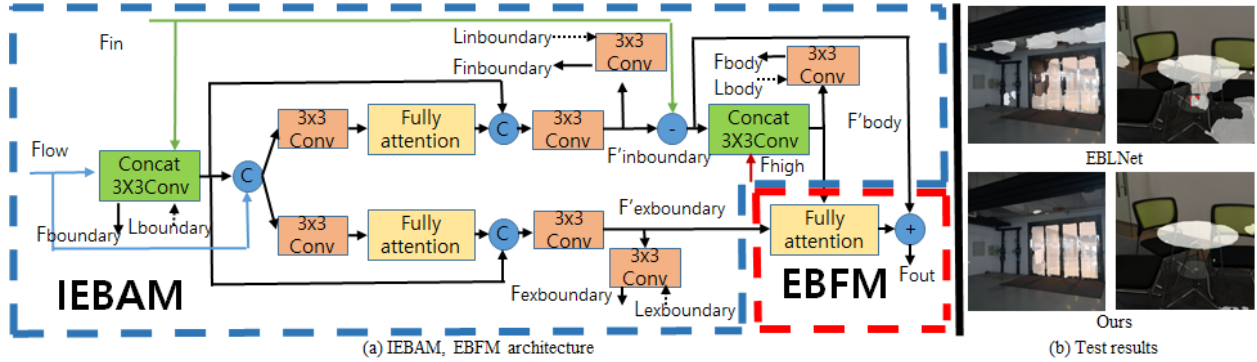


Figure 1: (a) Proposed architecture of our method, (b) Experimental comparison of segmentation results between our method and state of the art EBLNet: White highlighted area is transparent objects.

ABSTRACT

Recognizing transparent objects such as window and glassware is a challenging task in 3D Reconstruction from color images. In recent years, transparent object recognition methods have focused on feature extraction from boundary region of transparent objects. Our observation is that, unlike non-transparent objects, transparent objects can be characterized and located better by looking at their external and internal boundaries separately. We propose a new internal-external boundaries attention module in which internal and external boundary features are separately recognized. We add an edge-body fully attention module that supervises the segmentation of the generated transparent objects body using semantic information in the external boundaries. We employ contour loss to perform distance-weighted supervision on the inner and outer boundaries separately. Extensive experiments show that proposed method outperforms existing methods on Trans10k dataset.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; **Redundancy**; **Robotics**; • **Networks** → **Network reliability**.

KEYWORDS

glass, segmentation, boundary, module

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1 INTRODUCTION

Most of state of the art transparent glass object detection methods use boundary information to locate and segment glass region [He et al. 2021], [Xie et al. 2021]. However, these approaches do not consider the significant differences between external and internal region of the transparent object boundaries. Human recognize potential regions of transparent object from external boundary region such as window frame. Further investigation on internal boundary region determines whether window glass exist or not. Ignoring such semantic difference between internal and external boundary regions leads to incorrect segmentation of window frame without window glass or opened door. Based on the above observation, we propose Internal-External Boundary Attention Module (IEBAM), which utilizes contour loss [Chen et al. 2020] to guide the network to perceive internal and external boundaries of transparent objects and learn the distinction between the boundaries for accurate segmentation. Then we propose an Edge Body Fully-attention Module (EBFM), which utilizes the information of the external boundary features of transparent objects to supervise transparent body regions.

2 METHOD

The architectures of Internal External Boundaries Attention Module (IEBAM) and Edge Body Fully-attention Module (EBFM) are designed to provide step-by-step supervised guidance for neural

network training using different ground truths at different stages as shown in Fig. 1(a). The inputs to the module are F_{high}, F_{low} and F_{in} which are the same as EBLNet [He et al. 2021]. Our module replaces RDM module of EBLNet for supervised training.

2.1 Internal-External Boundaries Attention Module

Because lower layers of backbone networks contain more boundary information, we concatenate F_{in} and F_{low} and obtain boundary features $F_{boundary}$ by convolution operation. In order to be able to obtain the internal boundary features $F_{inboundary}$ accurately, we concatenate and use fully attention network [Song et al. 2022] to obtain internal boundary attentions. We concatenate it with the boundary features after convolution layer in eq.(1) which is Same for external boundaries.

$$F_{inboundary} = g_{3 \times 3}([f_{3 \times 3}([F_{boundary}; F_{low}]); F_{boundary}]), \quad (1)$$

where $[\cdot]$, $g_{3 \times 3}$ and $f_{3 \times 3}$ denote concatenation, convolution and fully attention network. F_{in} have transparent region features computed by the previous layer of the network. So we subtract the glass internal boundary $F_{inboundary}$ from F_{in} in order to obtain the glass body features. F_{high} contains a large amount of high receptive field information, so the glass body feature is concatenated with it and the final glass interior feature is obtained by convolution operation, in eq.(2):

$$F_{body} = g_{3 \times 3}([F_{high}; (F_{in} - F_{inboundary})]), \quad (2)$$

Finally, the $F_{inboundary}, F_{exboundary}, F_{body}$ were passed through the convolution operation and loss calculation with ground truth separately.

2.2 Edge Body Fully-attention Module

We utilize $F_{exboundary}$ as a fully-attention query feature, thus suppressing the feature map score of F_{body} the outside boundary belongs to the transparent object boundary, in eq.(3).

$$F_{body} = \gamma \sum_{i=1}^c \frac{\exp(Q_{exboundary(i)} \cdot K_{body(j)})}{\sum_{i=1}^c \exp(Q_{exboundary(i)} \cdot K_{body(j)})} + F_{body}, \quad (3)$$

where features K and Q and others are calculated in the same way as fully-attention, we just utilize $F_{exboundary}$ as input to calculate Q, and perform correlation and attention with F_{body} . Finally, we have $F_{out} = F_{body} + F_{inboundary}$. F_{out} is also transferred to the next layer and the graph convolutional network for training as in EBLNet.

2.3 Decomposed Contour Loss

Inspired by contour loss [Chen et al. 2020], we assign decomposed internal and external losses applying spatial weight maps in cross entropy loss. The weight map assigns relatively high value to near pixels, because closer pixels to transparent object boundaries contains richer semantic information.

3 RESULT AND DISCUSSION

We adopt same training settings as the original Trans10k [Xie et al. 2021] for fair comparison. We utilize pre-trained ResNet50 as backbone network and refine it with our new modules. We use 2 GPUs

Table 1: Comparison to state-of-the-art on Trans10k test set. All methods are trained on Trans10k training set under the same setting and all models use single scale inference.

Method	mIoU $\uparrow$	Acc $\uparrow$	mAE $\downarrow$	mBER $\downarrow$
Translab (ResNet50)	86.23	92.30	0.0081	6.76
EBLNet (ResNet50)	88.21	95.50	0.0073	5.97
Ours (ResNet50)	89.91	96.11	0.0061	5.28



(a)EBLNet



(b)Ours

Figure 2: Comparison test results of EBLNet and our method on pseudo-glass-like images in the Trans10k dataset. The white and red highlight areas in the result image are glass stuff and things respectively.

for all experiments and batch size of each GPU is set to 4. We report mIoU, mBER, mAE, ACC when comparing with other methods on test set. Segmentation result accuracy of EBLNet is the highest among the recently released glass-like and transparent object detection algorithms. We found that our model obtains higher accuracy than EBLNet with the same equipment and training parameters as shown in Table 1, especially on images such as chair legs and arches in the Trans10k dataset (in Fig. 2(b)).

In this work, we propose a new method for segmenting transparent objects. Compared with previous methods, our method improves the accuracy and reduces segmentation errors of transparent objects by extracting the internal and external boundary features of glass object and supervising glass segmentation. Proposed new method is proved on public dataset and we expect our approach can be utilized better in future research thereby improving the segmentation accuracy. the weight is promise

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